

Projeto de Licenciatura em Engenharia Biomédica
Edição 2025/2026 - 2º Semestre

Código:

(P000 - a preencher apenas pelo coordenador da disciplina)

Título:

AI4MDR | agentic-AI for Medical Device Regulation Certification

Orientador

Mário Zenha-Rela (mzrela@dei.uc.pt)

Número de alunos a que se destina a proposta:

2

Centro de Investigação/Empresa em que decorrerá o projeto

CISUC – Centro de Informática e Sistemas da UC (@DEI-UC Pólo II)

Descrição e Objetivos:

Any software developed for Medical Applications needs to be certified to be used in the European Union, as stated in the EU MDR directive. This regulation requires the adoption of the ISO/IEC 62304 standard for software development. In practice, this demands that every software development activity is recorded and traced along the development workflow. However, these processes are costly, slow, and error-prone due to its heavy dependence on manual tasks and extensive documentation. While efforts to (semi-automate) this process have been pursued, the resulting solutions are programmatically complex and fragile, making them susceptible to changes in the development ecosystem that jeopardize reliability.

*In this project we seek to explore alternative approaches, namely **assess the feasibility of adopting agent-AI** to address this problem, focused on the collection and generation of certification evidence.*

At the end of this project the students:

- Be knowledgeable of the EU MDR (Medical Device Regulation).
- Get a comprehensive understanding of the medical software development workflow.
- Know how to use AI-based software development tools (e.g. Claude Code, Cursor).
- Shall have a hands-on experience with agentic-AI software development.

Plano de Trabalhos:

The workplan has two distinct phases:

1. **Background (Feb):**
Learn the core competencies: the EU MDR; ISO/IEC 62304, SW Engineering, Agentic-AI.
2. **Iterative assessment and experimentation (Mar-May):**
Identify a scoped problem, design a solution, implement, test, assess and document.
Three problem-to-reporting cycles are expected (Mar/Apr/May).

Outras Observações:

All documentation shall be provided and expected in english.